# Dual-Band Microwave Sensor with Machine Learning for Milk Adulteration Detection

**Project report**

Course: BECE305L – Antenna and Microwave Engineering (Slot A2+tA2), VIT Chennai

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## 1. What we did

We built a small microwave sensor in CST that can show whether milk has been adulterated, what was added, and roughly how much. The sensor is a microstrip line on FR-4 with two square split-ring resonators of different size. Each ring shows up as a dip (notch) in the transmission curve S21. When a liquid sits over a ring, its permittivity shifts the dip, changes its depth and changes its sharpness (Q).

The steps were:

1. Start from published dual-band split-ring sensors (Wang et al., 2021; Mao and Liu, 2025) and build a two-ring design in CST.
2. Tune it with parameter sweeps (feed width, patch sizes p1 and p2, slot gap g2, ring spacing x2) and the optimiser.
3. Load a liquid cell on the ring with milk samples, using permittivity values taken from the literature.
4. Record the resonant frequency f1, the notch depth and the Q for 16 samples.
5. Train a machine-learning model on those readings, and wrap it in a Streamlit app that works in both directions.

## 2. Why it is different

In the papers we reviewed, most milk-adulteration sensors give one number, the frequency shift of a single resonator. That shows that something changed, but not what was added. Many are also validated on clean solvents rather than milk. Our work differs in these ways:

- **It names the adulterant, not only detects it.** We use three readings per sample (f1, notch depth, Q) and let a machine-learning model separate water, sugar and electrolyte-type adulteration.

- **It works in both directions.** Forward: enter a reading and get the adulterant and its amount. Inverse: choose an adulterant and an amount and get the expected f1, depth, Q and a synthesised S21 curve.
- **It uses milk-specific permittivity data** (milk, water-diluted milk, sugar-adulterated milk, synthetic milk) instead of pure solvents.
- **It reports its limits.** Validation is leave-one-out on a small dataset, and the weak points are stated in Section 9.

## 3. Sensor design

| Item | Value |
|---|---|
| Substrate | FR-4, εr = 4.4, thickness 1.6 mm, copper 0.035 mm |
| Ring 1 (lower band) | patch p1 = 9.76 mm square, slot gap 0.2 mm, liquid cell inner diameter 13 mm |
| Ring 2 (upper band) | patch p2 = 5.26 mm square, slot gap g2 = 0.2 mm, liquid cell inner diameter 12 mm |
| Ring spacing | x2 − x1 = 30.46 mm |
| Feed line | 1.6 mm wide, 50 Ω |
| Unloaded response | f1 = 2.168 GHz (−29.7 dB, Q ≈ 34); f2 ≈ 4.5 GHz (−23 dB) |

What the sweeps showed:

- p1 moves the lower notch (f1) and p2 moves the upper notch (f2), so each ring sets its own band.
- A larger g2 (0.3 mm) removes the upper notch.
- x2 shifts the upper notch.
- Feed width has only a weak effect on the notches.
- Our first design (DA1) looked better than it was (S21 near −45 dB) because of a coarse mesh and a small port. After refining the mesh and enlarging the port, the notches settled at about −29 and −23 dB.

## 4. Simulation setup and data

There are 16 samples: air (unloaded reference), fresh milk, synthetic milk, water added to milk (10, 20, 35, 50, 70, 85 and 100 ml), sugar added to milk (2, 5, 10 and 15 g) and a low and a high electrolyte mixture.

- Permittivity for milk, water, sugar and synthetic milk at 2.45 GHz comes from Malame et al. (2014).
- The electrolyte mixture comes from Eldamak et al., Biosensors (2020). It is a sweat-type mixture of several salts and urea, so it is **not pure urea**.
- Amounts that were not measured in the papers (for example water 35 ml, 70 ml, 85 ml, sugar 2 g and 15 g) are linear interpolations between the published points. Water 10 ml used an extrapolated value.
- We also tried an ethanol–water series, but its results did not follow permittivity, so we left it out.

## 5. Why only the first ring (about 2.2 GHz)

All readings in the dataset use the **first ring only**, at about 2.2 GHz. The reason is that published dielectric data for milk and its adulterants is plentiful around 2.45 GHz, which is the ISM band, and scarce near the second ring's 4.5 GHz. Using the first ring let us give every sample a literature-backed permittivity. The second ring is part of the design and was tuned and simulated unloaded, but it is not used in the dataset or the model. Using it is future work.

## 6. Machine learning and the app

- **Inputs:** f1, notch depth, Q and bandwidth.
- **Which adulterant:** a Random Forest classifier (100 trees).
- **How much:** a monotonic spline per adulterant that turns f1 into a concentration.
- **Inverse mode:** the same relations run backwards to give f1, depth, Q, bandwidth and a Lorentzian S21 curve.
- **App:** Streamlit, with forward and inverse tabs and a view of the classification confidence.

Validation was leave-one-out on the 16 samples, because the dataset is too small for a normal train/test split.

- Overall accuracy was 56.25 % (9 of 16 samples).
- On Water and Sugar, the two classes with enough samples, 9 of 11 samples were right (81.8 %). Sugar recall was 1.00 and Water recall 0.71.

- Classes with a single sample (air, fresh milk, synthetic milk) cannot be scored by leave-one-out, because the sample is missing from training when it is tested.
- Concentration error on the training points was about 3.8 ml for water and 1.4 g for sugar. A quick held-out check of a similar pipeline gave roughly 20 ml and 7.5 g, so reading concentration from f1 alone is only approximate. [Confirm this held-out figure with the team's own leave-one-out run.]

## 7. Results

| Sample | εr′ | f1 (GHz) | Depth (dB) | Q1 |
|---|---|---|---|---|
| Air (unloaded) | 1.00 | 2.1677 | −29.68 | 34.4 |
| Fresh milk | 69.55 | 1.9928 | −27.87 | 17.8 |
| Synthetic milk | 72.09 | 1.9832 | −19.55 | 6.8 |
| Water 10 ml | 73.03 | 1.9814 | −22.82 | 9.4 |
| Water 20 ml | 73.86 | 1.9590 | −22.82 | 9.4 |
| Water 35 ml | 75.11 | 1.9567 | −22.52 | 8.8 |
| Water 50 ml | 76.36 | 1.9110 | −22.58 | 10.0 |
| Water 70 ml | 76.97 | 1.9432 | −20.54 | 6.9 |
| Water 85 ml | 77.42 | 1.9300 | −21.99 | 8.3 |
| Water 100 ml | 77.88 | 1.9220 | −21.81 | 8.3 |
| Sugar 2 g | 62.96 | 2.1000 | −23.24 | 10.4 |
| Sugar 5 g | 63.05 | 2.1100 | −23.31 | 10.2 |
| Sugar 10 g | 63.20 | 2.1087 | −22.64 | 9.6 |
| Sugar 15 g | 63.35 | 2.0770 | −23.08 | 11.3 |
| Electrolyte mixture, low | 76.42 | 1.9366 | −21.31 | 7.8 |
| Electrolyte mixture, high | 73.96 | 1.9639 | −16.22 | 4.6 |

Main observations:

- Loading with milk lowers f1 from 2.168 GHz to about 1.99 GHz and halves the Q, because milk is lossy.

- Water lowers f1 further, by about 0.66 MHz for every ml added (1.981 to 1.922 GHz from 10 to 100 ml).
- Sugar leaves f1 close to the air value (about 2.08 to 2.11 GHz), so water and sugar move f1 in clearly different ways.
- A higher electrolyte level makes the notch shallower (−21.3 to −16.2 dB) and much broader.
- Normalised sensitivity is about 0.12 %/εr′ for milk, 0.15 for 100 ml water and 0.04 for 5 g sugar. This is lower than some published dual-band sensors (roughly 0.3 to 1.1 %/εr′). Our aim was to tell adulterants apart rather than to reach the highest sensitivity.

## 8. About the model files

All simulations were run in CST Studio Suite 2026, Education (student) edition. The original model we made in the lab is no longer available, so the CST model attached to this submission is the model we built for DA1 in the student version. The student version also limits the mesh size, so we ran with a coarse mesh and the results carry about ±0.02 to 0.03 GHz of simulation noise. That noise is as large as the whole sugar shift, which is one reason sugar concentration is hard to read.

The machine-learning code and app were written separately in Python (scikit-learn, SciPy, Streamlit) and run on the dataset above.

## 9. Limitations and next steps

- Everything is simulated. There is no VNA measurement of a fabricated sensor yet.
- The dataset is small (16 samples) and the permittivities come from the literature, with some interpolated.
- Only the first ring is used. The second ring is not used in the model.
- No published dielectric data was found for urea alone or for detergent, so those adulterants are not covered.
- Concentration estimates are rough; more samples at different amounts, run with identical mesh settings, would help.
- Next steps: fabricate the sensor on FR-4 and measure with a VNA, add the second ring to the model, and validate concentration on held-out samples.

## References

1. H. Mao and G. Liu, "Dual-Mode Microwave Sensor Based on BF-CSRR for Liquid Permittivity Measurement," IEEE Sensors Journal, vol. 25, no. 19, 2025.

2. C. Wang, X. Liu, L. Gan and Q. Cai, "A Dual-Band Non-destructive Dielectric Measurement Sensor Based on Complementary Split-Ring Resonator," Frontiers in Physics, vol. 9, 669707, 2021.
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4. Eldamak et al., "Study of the Dielectric Properties of Artificial Sweat Mixtures at Microwave Frequencies," Biosensors, vol. 10, no. 6, 2020.